

Reasoning with RAGged events: RAG-Enhanced Event Knowledge Base Construction and reasoning with proof-assistants

Stergios Chatzikyriakidis^[0000–0001–8016–9743]

Computational Linguistics and Language Technology lab
Department of Philology
University of Crete
`stergios.chatzikyriakidis@uoc.gr`

Abstract. Extracting structured computational representations of historical events from narrative text remains computationally expensive when constructed manually. While RDF/OWL reasoners enable graph-based reasoning, they are limited to fragments of first-order logic, preventing deeper temporal and semantic analysis. This paper addresses both challenges by developing automatic historical event extraction models using multiple LLMs (GPT-4, Claude, Llama 3.2) with three enhancement strategies: pure base generation, knowledge graph enhancement, and Retrieval-Augmented Generation (RAG). We conducted comprehensive evaluations using historical texts from Thucydides. Our findings reveal that enhancement strategies optimize different performance dimensions rather than providing universal improvements. For coverage and historical breadth, base generation achieves optimal performance with Claude and GPT-4 extracting comprehensive events. However, for precision, RAG enhancement improves coordinate accuracy and metadata completeness. Model architecture fundamentally determines enhancement sensitivity: larger models demonstrate robust baseline performance with incremental RAG improvements, while Llama 3.2 shows extreme variance from competitive performance to complete failure. We then developed an automated translation pipeline converting extracted RDF representations into Coq proof assistant specifications, enabling higher-order reasoning beyond RDF capabilities including multi-step causal verification, temporal arithmetic with BC dates, and formal proofs about historical causation. The Coq formalization validates that RAG-discovered event types represent legitimate domain-specific semantic structures rather than ontological violations. This work establishes that optimal enhancement strategy selection requires use-case-specific evaluation criteria rather than one-size-fits-all approaches, while demonstrating how automatic ontology construction can be grounded in formal verification for computational humanities applications.¹

Keywords: Historical Event Extraction · RAG · LLMs · Coq

1 Introduction

Ontologies play a crucial role in computational linguistics and artificial intelligence by codifying knowledge in a structured manner that enables sophisticated information retrieval and reasoning. These formal knowledge representations allow machines to understand semantic relationships between entities, facilitating complex queries that go beyond simple keyword matching. The DBpedia

¹ Code and experimental data are available at https://github.com/StergiosCha/RAGged_events.

ontology [19], with its 768 classes arranged in a subsumption hierarchy and described by over 3,000 properties, exemplifies the rich potential of such structured knowledge repositories for enhancing natural language understanding.

Despite their utility, structured knowledge extraction faces two fundamental challenges that have motivated extensive research in Retrieval-Augmented Generation (RAG). First, manual construction of event knowledge bases remains computationally expensive and labor-intensive. Second, existing RDF/OWL reasoning mechanisms are fundamentally limited to fragments of first-order logic, constraining their ability to handle complex temporal semantics, multi-step causal relationships, and sophisticated analytical queries essential for historical analysis.

While RAG has demonstrated success across various natural language processing tasks, its effectiveness for domain-specific knowledge extraction remains an open empirical question. The prevailing assumption that external knowledge enhancement universally improves performance may not hold when large language models already possess substantial internal knowledge about well-documented domains. Furthermore, the interaction between model architecture and enhancement strategies—particularly for specialized tasks like historical event extraction has received limited systematic investigation.

This paper presents a comprehensive empirical analysis that challenges conventional assumptions about RAG benefits through systematic comparison of automatic historical event knowledge base extraction across multiple LLM architectures and enhancement strategies. We investigate three distinct approaches, base generation without external knowledge, knowledge graph enhancement, and full RAG enhancement, using three state-of-the-art models: GPT-4o, Claude-3.5, and Llama 3.2. Our analysis reveals counterintuitive findings that fundamentally reshape understanding of when and how external knowledge augmentation should be applied. Our investigation uncovers an "inverse calibration principle" where enhancement effectiveness inversely correlates with model capability: stronger models like GPT-4o and Claude-3.5 achieve superior performance through pure base generation (extracting 36 and 39 comprehensive events respectively), while external enhancement often proves detrimental—introducing hallucinations, reducing coverage, and corrupting semantic accuracy. Conversely, weaker models like Llama 3.2 require external scaffolding but exhibit extreme sensitivity to implementation quality, ranging from competitive performance to catastrophic failure based on enhancement complexity.

These findings reveal that enhancement strategies optimize different performance dimensions rather than providing universal improvements. Base generation excels in comprehensive historical coverage and narrative richness, while RAG enhancement trades breadth for technical precision—improving coordinate accuracy and metadata completeness at the cost of event volume and semantic coherence. This trade-off suggests that optimal strategy selection must be driven by specific use-case requirements rather than blanket enhancement application.

To address the fundamental expressivity limitations of RDF/OWL systems revealed through our extraction analysis, we develop an automated translation pipeline that converts extracted knowledge representations into formal specifications for the Coq proof assistant. This translation enables higher-order reasoning capabilities impossible within traditional semantic web frameworks, including verification of multi-step causal relationships, complex temporal arithmetic with historical dates, and formal proofs about historical causation. The Coq formalization also validates a crucial theoretical finding: RAG-discovered event types that appear as ontological violations when evaluated against general-purpose knowledge graphs actually represent legitimate domain-specific semantic structures amenable to rigorous mathematical proof.

Our methodology employs historical texts from Thucydides’ History of the Peloponnesian War as a controlled domain for systematic evaluation. This choice enables assessment of enhancement strategies against well-documented historical events while revealing how model architecture and domain characteristics interact to determine optimal knowledge integration approaches. The integration of automatic extraction with formal verification establishes a foundation for computational humanities applications that combine modern NLP scalability with mathematical rigor. The central contribution of this work is demonstrating that enhancement strategy effectiveness depends fundamentally on the interaction between model capability, domain characteristics, and intended application requirements, challenging the field’s assumption that more comprehensive external knowledge necessarily leads to better performance.

2 Related work

Retrieval-Augmented Generation (RAG) is an approach that combines the strengths of retrieval-based systems, which fetch relevant information from external sources, and generative models, which synthesize this extracted information into coherent and contextually aware outputs [20]. RAG enables large language models (LLMs) to dynamically access up-to-date and domain-specific knowledge, thus improving their factual accuracy, reducing hallucinations, and supporting more nuanced reasoning [13].

RAG models have achieved state-of-the-art results across various tasks, including open-domain question answering, information retrieval, and knowledge-grounded text generation [18]. By leveraging external memory (such as Wikipedia or specialized knowledge bases), RAG architectures outperform purely parametric models, particularly in generating more specific, diverse, and factually grounded responses [16]. The typical RAG workflow involves retrieving relevant documents or passages, fusing this information with the input prompt, and then generating the final output using an LLM [20].

Beyond general question answering, RAG techniques have been successfully applied to information extraction (IE) tasks, including named entity recognition, relation extraction, and event extraction [34]. In these settings, RAG enhances both performance and interpretability by grounding extractions in retrieved, contextually relevant evidence. For instance, in named entity recognition, retrieval of similar sentences improves semantic representations, while in event argument extraction, retrieval-augmented models help capture cross-argument dependencies and improve few-shot learning performance [21].

Recent studies have specifically explored RAG for event argument extraction, a task that involves identifying and classifying the participants and attributes of events described in text [10]. Retrieval-augmented generative models, such as R-GQA [11] and DRAGEAE [14], retrieve similar QA pairs or demonstration examples to inform the generation process, resulting in substantial improvements over extractive and standard generative baselines. These approaches are particularly effective in scenarios with limited annotated data, as they can leverage demonstrations from related contexts to guide extraction [25].

RAG has been used in temporal event relation extraction to address the inherent ambiguity of temporal semantics in text [37]. By retrieving relevant knowledge from LLMs and integrating it into prompt templates, RAG-based methods achieve superior performance in extracting and reasoning about event temporal relations [23].

Another relevant line of work is grounding RAG in structured ontologies, for example domain-specific knowledge graphs [2]. Ontology-Grounded RAG (OG-RAG) approaches basically construct

hypergraph representations of domain documents and cluster factual knowledge according to ontological relationships [7,24].

Recent comparative studies of RAG configurations reveal that the relationship between retrieval complexity and performance is non-linear [9]. Our work contributes to this emerging understanding by demonstrating that for domain-specific knowledge extraction, particularly in well-documented historical domains, moderate retrieval strategies can outperform both minimal and maximal external knowledge integration [26]. This challenges the prevalent assumption in the RAG literature that more comprehensive retrieval necessarily leads to better performance, highlighting the importance of domain-specific optimization in RAG system design [1].

Lastly, Coq has been used to formalize natural language temporal and aspectual semantics by [6] as part of a symbolic Natural Language Inference system, achieving state-of-the-art results for symbolic approaches. In general, Coq has been shown to be a powerful tool to formalize natural language semantics [8].

Our proposal has similarities with some of the related work, but it is unique in its main approach. What we offer is twofold: a) the use of RAG for automatic event knowledge base extraction with systematic comparison of different RAG and non-RAG configurations, and b) a translation of the resulting extracted events into the Coq proof assistant, enabling advanced temporal and causal reasoning beyond what is possible with simple RDF languages [15,28].

3 Methodology

The proposed approach implements a two-phase pipeline for transforming unstructured historical narratives into formal knowledge representations. Phase 1 employs a number of NLP techniques for semantic event extraction, while Phase 2 performs logic translation to enable automated reasoning and formal verification:

- **Phase 1**
 - Event boundary detection and segmentation
 - Agent identification with semantic role assignment
 - Geographical entity resolution and coordinate mapping
 - Temporal expression normalization
 - Outcome extraction and causal relationship identification
 - RDF knowledge graph construction
- **Phase 2**
 - RDF-to-Coq inductive type conversion
 - Higher-order temporal logic implementation
 - Causal inference framework integration
 - Proof-assistant compatibility for formal verification

In phase 1, we use three enhancement strategies across three LLM architectures using historical text from Thucydides’ *History of the Peloponnesian War* (first 9 chapters). Our knowledge sources include the DBpedia ontology that involves 768 classes and 3,000+ properties, and the specialized LACRIMALit ontology) for Ancient Greek geographical locations [27]. External knowledge retrieval includes the Wikidata and the DBpedia SPARQL endpoints, as well as the ConceptNet API. The three models we use are GPT-4o (large commercial model), Claude-3.5 Sonnet (alternative large-scale architecture), and Llama 3.2 (3.2B parameter model via local Ollama deployment). All models

use deterministic settings (temperature=0) for consistency and use the same chunking parameters. We use three variations on these models, base-generation, generation with knowledge enhancement, and generatoin with knowledge enhancement and RAG.

True Base Generation (Control): Pure LLM inference without external knowledge augmentation. All APIs, knowledge graphs, location enrichment, and RAG functionality are explicitly disabled for baseline measurement. However, the baseline uses the structured prompt shown in figure 1.

System Prompt for Historical Event Extraction

You are extracting historical events from text using **ONLY** the information provided in the text chunk.

TEXT CHUNK {chunk_num} TO ANALYZE:
{chunk}

ENTITIES FOUND: {entities}
LOCATIONS FOUND: {locations}

TASK: Extract historical events using text information and making REASONABLE INFERENCES.

REQUIREMENTS:

1. Extract **ONLY** events explicitly mentioned in the text chunk
2. Use information directly stated in the text
3. **MAKE REASONABLE INFERENCES** from context clues
4. Infer coordinates, countries, regions from textual context

OUTPUT FORMAT:

```
ste:Event{chunk_num}_1 a ste:Event ;
ste:hasType "event description" ;
ste:hasAgent "person/group" ;
ste:hasTime "time period" ;
ste:hasLocation "location name" ;
ste:hasLatitude "37.9838" ;
ste:hasLongitude "23.7275" ;
ste:hasCountry "Greece" ;
ste:hasRegion "Attica" ;
ste:hasLocationSource "inferred" ;
ste:hasResult "outcome" .
```

Fig. 1. LLM System Prompt for Historical Event Extraction

Knowledge Enhanced (No-RAG): Here, we use knowledge integration but no RAG. We attempt entity extraction, knowledge graph retrieval queries to Wikidata, DBpedia, and ConceptNet.

Lastly, we attempt location enrichment that follows the order LACRIMALit ontology $\rightarrow$ Wikidata $\rightarrow$ DBpedia, stopping upon successful coordinate retrieval.

RAG Enhancement: This adds a Retrieval-augmented generation layer that combines vector similarity with knowledge graph enhancement. Implementation uses FAISS vector store with HuggingFace paraphrase-multilingual-MiniLM-L12-v2 embeddings, and different parameters for k . The RAG pipeline is shown in figure 2.

3.1 Implementation and Evaluation

Our system employs cascading queries (local ontology $\rightarrow$ Wikidata $\rightarrow$ DBpedia) with MD5-based caching and rate limiting to minimize API calls. Evaluation combines quantitative metrics (event coverage, coordinate precision, RDF completeness) with qualitative assessment of semantic depth and ontological classification. All outputs follow standardized RDF/Turtle format using consistent properties: `ste:hasType`, `ste:hasAgent`, `ste:hasTime`, `ste:hasLocation`, `ste:hasLatitude/Longitude`, and `ste:hasResult`. `ste:hasType`, `ste:hasAgent`, `ste:hasTime`, `ste:hasLocation`, `ste:hasLatitude/Longitude`, and `ste:hasResult` to enable direct comparison across enhancement strategies.

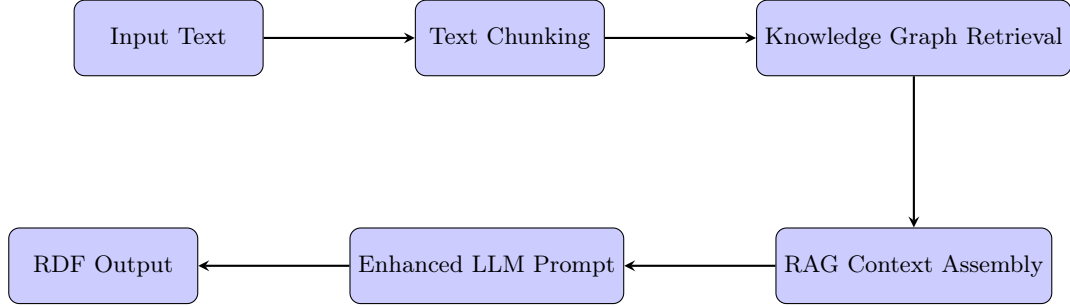


Fig. 2. The RAG Pipeline

3.2 Enhancement Strategy Comparison

Our systematic investigation examined how different knowledge integration approaches affect extraction performance across model architectures. We evaluated the three enhancement strategies using identical prompting frameworks to ensure fair comparison. The results are shown in Table 1. Our analysis reveals distinct performance trade-offs that challenge conventional assumptions about external knowledge benefits. The results demonstrate that enhancement strategies are detrimental with stronger models and helpful with less powerful models. Base generation consistently achieved maximum event coverage (GPT-4: 36 events, Claude: 39 events) with rich narrative detail and comprehensive historical scope, producing almost clean RDFs in terms of syntax (the only thing missing is xsd format, that was missing in the prompt for base but included in the other architectures). Knowledge-base and RAG consistently produced fewer events, while at the same time introduced hallucinations into the text. For example, the coordinates for the battle of Sybota are correctly attributed to Sybota in base GPT-4o, but incorrectly given in Knowledge Enhancement

GPT-4o. A bit of research reveals that the coordinates used belong to Ambracia, one of the cities that participated in the battle of Sybota. Claude-RAG conflates three distinct temporal layers: ancient events (431 BC battles, sieges), authorship (Thucydides writing about the war), and modern publishing (Crawley translations, Project Gutenberg updates), extracting all as if they were contemporary Peloponnesian War events. Additionally includes metadata like *hasLLM: Claude4* as RDF properties.

LLAMA exhibits the most dramatic variation in performance, revealing a brittle failure mode under complex enhancement conditions. Base LLAMA produces only 10 events with poor formatting, establishing a low baseline. Simple RAG configurations (RAG 1-2) deliver substantial improvements, boosting output to 37-38 events with better structure, demonstrating that knowledge augmentation can effectively compensate for weaker model capabilities. However, this improvement exhibits catastrophic failure at higher complexity levels: more sophisticated RAG setups cause complete breakdown, with RAG 3 producing only 6 events and RAG 4 achieving zero correct event structures. This suggests that weaker models benefit from controlled knowledge injection but lack the architectural robustness to handle complex multi-source reasoning without systematic collapse.

These findings reveal a nuanced inverse calibration principle that operates along two critical dimensions: enhancement intensity and model capability. Recent empirical work has demonstrated that stronger retrievers paradoxically cause more performance degradation, with performance exhibiting an "inverted-U pattern" where initial improvements are followed by sharp decline [17]. Our results extend this phenomenon beyond retrieval to encompass broader knowledge integration strategies, revealing that the relationship between model strength and enhancement benefit follows a complex inverse calibration where stronger models perform optimally with minimal augmentation while weaker models require carefully bounded enhancement.

The LLAMA trajectory, i.e. from baseline failure (10 events) through optimal performance (37-38 events) to catastrophic collapse (0 events), illustrates a fragility threshold in knowledge integration. This aligns with findings that weaker models are more susceptible to "hard negatives" and can only handle effective knowledge augmentation up to specific complexity limits before experiencing systematic breakdown [17]. The improvement from base to simple RAG demonstrates that weaker architectures can benefit substantially from controlled knowledge injection, effectively compensating for knowledge deficits. However, the catastrophic failure at higher RAG complexity levels reveals that these same models lack the architectural robustness to handle multi-source reasoning without semantic collapse.

Conversely, stronger models like GPT-4o and Claude exhibit the opposite pattern: their sophisticated parametric knowledge and reasoning capabilities are disrupted rather than enhanced by external knowledge sources. This supports research showing that stronger models exhibit different relationships between internal knowledge and retrieved context, often prioritizing external information even when it contradicts their more accurate training data [32]. The coordinate misattribution in GPT-4o and temporal conflation in Claude-RAG exemplify how knowledge augmentation can introduce noise that overwhelms these models' pattern recognition capabilities, causing them to move away from their reliable internal representations in favor of potentially corrupted external sources. This phenomenon reflects the theoretical framework where knowledge integration increases cognitive load and introduces semantic drift, particularly problematic for models with sophisticated internal knowledge structures that can be destabilized by conflicting external information [12].

Beyond performance metrics, RAG in some cases also appears to corrupt the semantic accuracy of extracted events and disrupt ontological consistency when evaluated against standard knowledge graphs. While some extracted events align with DBpedia subtypes (e.g., SocietalEvent), most

Model	Configuration	Events	Precision	RDF	Issues
GPT	Base Generation	36	Excellent	Excellent	Missing xsd (due to prompt)
GPT	No RAG + KG	30	Good (some Hallucinations)	Excellent	Some empty coordinates
GPT	RAG + Multi KG 1	23	Good (some Hallucinations)	Excellent	Some empty coordinates
GPT	RAG + Multi KG 2	23	Good (some Hallucinations)	Excellent	some empty coordinates
GPT	RAG + Multi KG 3	20	Good (some Hallucinations)	Perfect	No syntax issues
GPT	RAG + Multi KG 4	26	Good (some Hallucinations)	Perfect	No syntax issues
Claude 4	Base Generation	39	Excellent	Excellent	Missing xsd (due to prompt)
Claude 4	No RAG + KG	20	Fair (hallucinations)	Good	Repeats event ids
Claude 4	RAG + Multi KG 1	10	Poor (Many Hallucinations)	Good	Missing coordinate fields
Claude 4	RAG + Multi KG 2	16	Poor (Many Hallucinations)	Good	Missing coordinate fields
Claude 4	RAG + Multi KG 3	19	Poor (Many Hallucinations)	Good	Missing coordinate fields
Claude 4	RAG + Multi KG 4	19	Poor (Many Hallucinations)	Good	Missing coordinate fields
LLAMA	Base Generation	10	Fair	Poor	Many formatting errors
LLAMA	No RAG + KG	19	Good	Fair	Some formatting errors
LLAMA	RAG + Multi KG 1	38	Good	Good	Some missing coordinates and repeated event ids
LLAMA	RAG + Multi KG 2	37	Good	Good	Some missing coordinates and repeated event ids
LLAMA	RAG + Multi KG 3	6	Poor	Poor	Many syntax errors
LLAMA	RAG + Multi KG 4	0	Bad	Bad	No correct event structure

Table 1. Model Performance Comparison

domain-specific event descriptions fall outside DBpedia’s taxonomy (e.g., there is no WarEvent or MigrationEvent in the standard ontology). The prompt’s reference to *dbp:SpecificEventType* reveals this ontological gap, i.e. models frequently default to this generic classification when encountering historically meaningful events that require finer semantic distinctions. However, these apparent inconsistencies reflect a fundamental granularity mismatch rather than genuine semantic errors. RAG consistently discovers domain-specific event types and relationships that exceed general-purpose ontologies’ coverage. While these extractions violate DBpedia’s constraints, they often capture historically accurate and semantically coherent distinctions essential for domain analysis. This subevent discovery would be helpful when we will discuss our translation procedure to Coq.

Lastly, other cases where RAG messes up is cases where a broader historical context from the text is taken into consideration but the information presented is conflated and inaccurate like you see in the example below:

The example in table 2 illustrates not only the temporal conflation problem where Thucydides (the historian who documented the war) is incorrectly identified as the agent causing the Peloponnesian War itself, but also incorrectly identifies the war as *the greatest movement in history involving the Hellenes and large part of the barbarian world*. This clearly demonstrates how RAG enhancement can fundamentally corrupt the semantic relationships between historical actors and events instead of make them more precise and enriching them.

4 Beyond RDF: Enabling Higher-Order Reasoning

While our event extraction system successfully produces structured RDF knowledge graphs from historical texts, these representations encounter fundamental computational barriers when subjected to sophisticated analytical demands. The intricate temporal dependencies, multi-layered causal networks, and emergent narrative patterns inherent in historical discourse require reasoning architectures that transcend the computational boundaries of conventional semantic web frameworks.

Table 2. RDF Event Extraction Example: Temporal Conflation in Claude-RAG

Property	Value
<code>rdf:type</code>	<code>ste:Event, dbp:SpecificEventType</code>
<code>ste:hasType</code>	The outbreak of the Peloponnesian War
<code>ste:hasAgent</code>	Thucydides
<code>ste:hasTime</code>	431 BC
<code>ste:hasLocation</code>	Greece
<code>ste:hasLatitude</code>	39.0742 (xsd:double)
<code>ste:hasLongitude</code>	21.8243 (xsd:double)
<code>ste:hasCountry</code>	Greece
<code>ste:hasRegion</code>	Hellas
<code>ste:hasLocationSource</code>	wikidata
<code>ste:hasResult</code>	The greatest movement yet known in history, involving the Hellenes and a large part of the barbarian world
<code>ste:hasRAGContext</code>	yes

4.1 Computational Limitations of RDF/OWL Systems

Contemporary RDF/OWL reasoning engines operate within restrictive logical fragments that prove systematically inadequate for historical analysis. These systems are constrained to decidable subsets of first-order logic, fundamentally limiting their capacity to express and verify complex historical relationships. Consider the cascading causal sequence linking the Siege of Plataea (429-427 BCE) to Pericles’ political decline: the prolonged siege concentrated Athenian populations within city walls, creating conditions that amplified plague transmission, which subsequently shifted political sentiment and undermined Pericles’ leadership over multiple years. While RDF can represent these as discrete events with simple causal predicates (`:causedBy`), it cannot formally express the multi-step temporal dependencies or verify the logical consistency of such cascading relationships.

Traditional SPARQL queries lack the expressive power for inductive reasoning patterns essential to historical analysis. Questions such as “Do successful military campaigns correlate with subsequent political stability?” require pattern matching across temporal sequences and statistical inference over event distributions, capabilities that exceed SPARQL’s graph pattern matching paradigm. Furthermore, counterfactual reasoning demands modal logic constructs entirely absent from RDF/OWL specifications.

4.2 Formal Translation Architecture

We develop an automated translation pipeline that converts extracted RDF/Turtle representations into formal specifications for the Coq proof assistant. This architecture enables higher-order reasoning while preserving the semantic richness captured during RAG-enhanced extraction, validating our earlier findings about ontological discovery versus hallucination.

The translation system comprises four integrated components: (1) an RDF parsing module that extracts event structures, temporal relationships, and participant roles from Turtle serializations; (2) a type inference engine that maps RAG-discovered event classes to formal Coq inductive types, demonstrating that fine-grained classifications like `MigrationEvent` and `WarEvent` represent legitimate semantic structures rather than ontological violations; (3) a temporal logic translator that

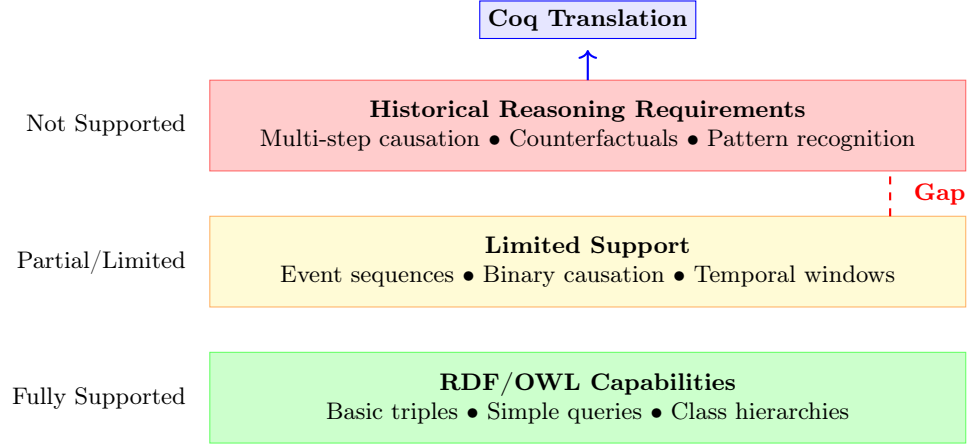


Fig. 3. RDF/OWL capabilities versus historical reasoning requirements. The reasoning gap between limited RDF expressivity and complex historical analysis necessitates translation to formal verification systems.

converts chronological relationships into mathematical constraints using dependent types; and (4) a theorem generation system that produces verifiable propositions about historical causation, enabling formal verification of complex historical hypotheses.

4.3 Higher-Order Reasoning Capabilities

The Coq formalization unlocks analytical capabilities impossible within RDF frameworks. Multi-step causal reasoning becomes expressible through function composition and dependent types:

```

Definition cascading_causation (e1 e2 e3 : Event) : Prop :=
  temporal_precedes e1 e2 /\ may_have_caused e1 e2 /\
  temporal_precedes e2 e3 /\ may_have_caused e2 e3 /\
  same_historical_context e1 e3.

```

This enables formal verification of historical propositions that require transitive reasoning across multiple temporal periods. Inductive definitions allow pattern recognition across event sequences, while dependent types capture contextual constraints that ensure logical consistency. The system can formally prove or disprove complex historical claims, transforming historical analysis from interpretive discourse into rigorous logical investigation.

Crucially, this translation validates our ontological discovery thesis: RAG-generated event types that violate DBpedia constraints integrate seamlessly into Coq’s rich type system, demonstrating their semantic legitimacy. The apparent ontological inconsistencies represent valuable domain-specific knowledge structures that formal reasoning systems can exploit for sophisticated historical analysis.

4.4 Translation Process

The automated translation from RDF/Turtle representations to Coq specifications operates through a four-stage pipeline that systematically transforms extracted event data into formal logical constructs while validating ontological discoveries.

Stage 1: Ontological Discovery Validation. The translation process begins by analyzing the class hierarchy extracted from RDF triples, distinguishing between standard DBpedia classes and RAG-discovered event types. Our validation algorithm assesses each non-standard class using semantic indicators (presence of domain-specific terms like “War”, “Migration”, “Siege”) and contextual coherence (temporal consistency, specific participants, geographical anchoring). This validation step demonstrates that apparent ontological violations often represent legitimate domain knowledge rather than hallucinations—for instance, `ET_MigrationEvent` represents historically accurate fine-grained event classification that exceeds DBpedia’s granularity.

Stage 2: Type System Construction. The translator constructs Coq’s rich type system by mapping RDF classes to inductive types with formal subclass relationships. RAG-discovered event types integrate seamlessly into this hierarchy, validating our thesis that these represent coherent semantic structures. The system generates dependent types for temporal reasoning, enabling formal arithmetic on BC dates through Coq’s `Z` library:

```
Definition TimePoint := (Z * nat * nat)%type.
Definition is_bc_event (e : Event) : bool :=
  match time e with
  | Some (y, _, _) => Z.ltb y 0%Z
  | None => false
end.
```

Stage 3: Relationship Formalization. Complex event relationships that exceed RDF’s expressive capacity become formally verifiable propositions. Multi-step causal reasoning emerges through function composition and dependent types, enabling verification of cascading historical causation impossible within SPARQL’s pattern matching paradigm:

```
Definition cascading_causation (e1 e2 e3 : Event) : Prop :=
  event_before e1 e2 = true /\ may_have_caused e1 e2 = true /\
  event_before e2 e3 = true /\ may_have_caused e2 e3 = true /\
  same_historical_context e1 e3 = true.
```

Stage 4: Theorem Generation. The final stage generates formal theorems that validate both the semantic coherence of RAG discoveries and enable complex historical reasoning. The `rag_discoveries_coherent` theorem formally proves that RAG-extracted events with proper temporal, participant, and class structure represent valid domain knowledge, while the another theorem demonstrates how domain-specific classifications enhance causal inference capabilities.

This translation architecture validates our central claim: RAG-discovered event types that appear as ontological violations in RDF evaluation can potentially represent valuable semantic innovations that formal reasoning systems can exploit for sophisticated historical analysis beyond the capabilities of traditional semantic web frameworks.

4.5 Toward Practical Applications: Foundation for Future Development

The Coq translation establishes a foundation for applications that would be impossible with traditional RDF/OWL systems, though significant development remains before practical deployment. Our system demonstrates the feasibility of formally representing complex historical relationships and verifying basic logical consistency within extracted event structures. While current applications

require Coq expertise, the architecture proves that RAG-discovered ontologies can support formal reasoning about historical causation.

The primary contribution lies in demonstrating that apparent RAG “hallucinations” represent legitimate ontological discoveries amenable to formal verification. This reframes evaluation criteria from conformity to existing knowledge graphs toward recognizing domain-specific semantic innovations. Rather than dismissing fine-grained event classifications as errors, our approach validates their semantic coherence through formal mathematical proof.

The translation architecture opens several promising research avenues. Future work could develop domain-specific logical frameworks that capture historical reasoning patterns without requiring full Coq proficiency. Automated consistency checking tools for digital humanities scholars represent another promising direction, as do hybrid systems that combine statistical correlation with formal logical constraints for causal analysis. Additionally, user-friendly interfaces could eventually expose formal verification capabilities to historians without formal logic training.

Several limitations constrain immediate practical deployment. The approach requires substantial formal logic expertise, limiting accessibility for working historians. The system currently handles relatively simple historical relationships, and scaling to complex historical narratives with multiple conflicting sources remains challenging. The translation process also requires manual verification of generated Coq definitions to ensure correctness.

This integration of automatic extraction with formal verification represents a foundational step toward rigorous computational humanities tools. While not immediately deployable for working historians, the methodology demonstrates that modern NLP techniques can be grounded in formal logical systems, suggesting promising directions for future tool development that could eventually provide both scalability and mathematical rigor for historical analysis.

5 Conclusions

Our systematic comparison of RAG configurations reveals a fundamental insight: optimal RAG design challenges conventional assumptions by demonstrating that base generation without external enhancement often outperforms both moderate and maximal retrieval strategies. The discovery that pure inferential generation achieves superior overall performance compared to enhanced RAG configurations, particularly for well-documented domains, has significant implications for the field.

These findings challenge the prevailing assumption that more comprehensive retrieval necessarily leads to better performance. Our analysis demonstrates that enhancement strategies optimize different performance dimensions rather than providing universal improvements. Stronger models like GPT-4o and Claude-3.5 demonstrate superior performance in pure base generation, producing comprehensive historical coverage with rich semantic detail, while external enhancement provides complementary but often detrimental effects: improved technical precision at the cost of reduced event coverage and introduction of hallucinations.

The architectural enhancement divide we identified suggests that model capacity fundamentally determines both baseline performance and enhancement sensitivity. Larger, extensively trained models possess robust internal knowledge representations that benefit selectively from external enhancement while potentially sacrificing narrative breadth. Smaller models like Llama 3.2 require external knowledge scaffolding but exhibit catastrophic failure modes when enhancement complexity exceeds their architectural robustness, demonstrating high variance from competitive performance to complete semantic breakdown.

Beyond performance metrics, our work reveals that RAG systems perform ontological discovery rather than mere information retrieval. The fine-grained event types that appear as violations when evaluated against general-purpose ontologies like DBpedia can actually represent legitimate domain-specific semantic structures. Events such as `ET_Revolt` or `ET_MigrationEvent` exceed DBpedia’s granularity while maintaining historical accuracy and semantic coherence. This reframes the evaluation paradigm from conformity to existing knowledge graphs toward recognizing domain-specific innovations.

Our translation to Coq demonstrates that these RAG-discovered event knowledge bases can support sophisticated logical reasoning impossible within traditional semantic web frameworks. The ability to express and verify multi-step causal relationships, perform complex temporal arithmetic with BC dates, and conduct formal proofs about historical causation represents a significant advancement beyond RDF/OWL capabilities. The successful integration of RAG-discovered event types into Coq’s rich type system validates their semantic legitimacy through mathematical proof.

The methodology establishes a foundation for computational humanities applications that combine the scalability of modern NLP techniques with the rigor of formal logical systems. While current implementation requires substantial formal logic expertise, the architecture proves that automatic ontology construction can be grounded in mathematical verification, opening promising directions for future tool development.

These findings suggest that RAG system design should prioritize careful evaluation of whether external enhancement is necessary, with configuration choices informed by domain-specific characteristics and model capabilities. For classical historical texts, the LLM’s built-in knowledge proves remarkably comprehensive, suggesting that the value proposition of external enhancement requires careful domain-specific evaluation rather than blanket application.

Future work should explore generalization across domains and historical periods, investigate hybrid approaches combining base generation’s comprehensive coverage with targeted enhancement’s precision, and develop accessible interfaces for formal verification. The integration of ontological discovery validation with automated reasoning offers promising directions for computational linguistics and digital humanities.

Acknowledgements

The author gratefully acknowledges funding from Amazon (project: Neural-Symbolic Integration for Enhanced Natural Language Processing (NIELS)).

References

1. Akari Asai, Sewon Min, Zexuan Zhong, and Danqi Chen. Retrieval-based language models and applications. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics: Tutorial Abstracts*, pages 1–50, 2023.
2. Jinheon Baek, Alham Fikri Aji, and Amir Saffari. Knowledge-Augmented Language Model Prompting for Zero-Shot Knowledge Graph Question Answering. In *Proceedings of the First Workshop on Matching From Unstructured and Structured Data (MATCHING 2023)*, pages 70–98, Toronto, ON, Canada, 2023. Association for Computational Linguistics.
3. R. C. Barron, V. Grantcharov, S. Wanna, M. E. Eren, M. Bhattarai, N. Solovyev, G. Tompkins, C. Nicholas, K. Ø. Rasmussen, C. Matuszek, et al. Domain-Specific Retrieval-Augmented Generation Using Vector Stores, Knowledge Graphs, and Tensor Factorization. In *2024 International Conference on Machine Learning and Applications (ICMLA)*, pages 1669–1676, 2024.

4. Bernardy, J.-P., Chatzikiyiakidis, S.: Applied temporal analysis: A complete run of the FraCaS test suite. In: *Proceedings of the 14th International Conference on Computational Semantics (IWCS)*. pp. 11–20 (2021)
5. Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego de Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, Jack W. Rae, Erich Elsen, and Laurent Sifre. Improving language models by retrieving from trillions of tokens. In *Proceedings of the 39th International Conference on Machine Learning*, pages 2206–2240, 2022.
6. Jean-Philippe Bernardy and Stergios Chatzikiyiakidis. Applied temporal analysis: A complete run of the FraCaS test suite. In *Proceedings of the 14th International Conference on Computational Semantics (IWCS)*, pages 11–20, 2021.
7. Kartik Sharma, Peeyush Kumar, and Yunqing Li. Og-rag: Ontology-grounded retrieval-augmented generation for large language models. *arXiv preprint arXiv:2412.15235*, 2024.
8. Stergios Chatzikiyiakidis and Zhaohui Luo. Proof assistants for natural language semantics. In *Logical Aspects of Computational Linguistics. Celebrating 20 Years of LACL (1996–2016) 9th International Conference, LACL 2016, Nancy, France, December 5–7, 2016, Proceedings 9*, pages 85–98. Springer, 2016.
9. Florin Cuconasu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonello, and Fabrizio Silvestri. The power of noise: Redefining retrieval for RAG systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 719–729, 2024.
10. Xinya Du and Claire Cardie. Event extraction by answering (almost) natural questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 671–683, 2020.
11. Xinya Du and Heng Ji. Retrieval-augmented generative question answering for event argument extraction. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 4649–4666, 2022.
12. Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.
13. Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023.
14. Shiming He, Yu Hong, Shuai Yang, Jianmin Yao, and Guodong Zhou. Demonstration retrieval-augmented generative event argument extraction. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 4617–4625, 2024.
15. Aidan Hogan, Eva Blomqvist, Michael Cochez, Claudia d’Amato, Gerard de Melo, Claudio Gutierrez, Sabrina Kirrane, José Emilio Labra Gayo, Roberto Navigli, Sebastian Neumaier, Axel-Cyrille Ngonga Ngomo, Axel Polleres, Sabbir M. Rashid, Anisa Rula, Lukas Schmelzeisen, Juan Sequeda, Steffen Staab, and Antoine Zimmermann. Knowledge graphs. *ACM Computing Surveys*, 54(4):1–37, 2021.
16. Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 874–880, 2021.
17. Bowen Jin, Jinsung Yoon, Jiawei Han, and Serkan O. Arik. Long-Context LLMs Meet RAG: Overcoming Challenges for Long Inputs in RAG. *arXiv preprint arXiv:2410*, 2024.
18. Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, 2020.

19. J. Lehmann, R. Isele, M. Jakob, A. Jentzsch, D. Kontokostas, P. N. Mendes, S. Hellmann, M. Morsey, P. Van Kleef, S. Auer, et al. DBpedia—a large-scale, multilingual knowledge base extracted from wikipedia. *Semantic Web*, 6(2):167–195, 2015.
20. Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.
21. Hao Li, Yanan Cao, Yubing Ren, Fang Fang, Lanxue Zhang, Yingjie Li, and Shi Wang. Intra-Event and Inter-Event Dependency-Aware Graph Network for Event Argument Extraction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6362–6372, Singapore, 2023. Association for Computational Linguistics.
22. Jasper Linders and Jakub M. Tomczak. Knowledge graph-extended retrieval augmented generation for question answering. *arXiv preprint arXiv:2504.08893*, 2025.
23. Jiawei Liu, Chunyu Kit, and Gina-Anne Levow. RAG-based temporal reasoning for event relation extraction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 13245–13253, 2023.
24. Haoran Luo, Guanting Chen, Yandan Zheng, Xiaobao Wu, Yikai Guo, Qika Lin, Yu Feng, Zemin Kuang, Meina Song, Yifan Zhu, and others. HyperGraphRAG: Retrieval-Augmented Generation with Hypergraph-Structured Knowledge Representation. *arXiv preprint arXiv:2503*, 2025.
25. Yubo Ma, Zehao Wang, Yixin Cao, and Aixin Sun. Few-shot event detection: An empirical study and a unified view. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 2845–2857, 2023.
26. Oded Ovadia, Menachem Brief, Moshik Mishaelli, and Oren Elisha. Fine-tuning or retrieval? comparing knowledge injection in llms. *arXiv preprint arXiv:2312.05934*, 2023.
27. Maria Papadopoulou, Christophe Roche, and Eleni-Melina Tamiolaki. The LACRIMALit Ontology of Crisis: An Event-Centric Model for Digital History. *Information*, 13(8):398, 2022.
28. Heiko Paulheim. Knowledge graph refinement: A survey of approaches and evaluation methods. *Semantic Web*, 8(3):489–508, 2017.
29. W. Su, Y. Tang, Q. Ai, Z. Wu, and Y. Liu. DRAGIN: Dynamic Retrieval Augmented Generation based on the Information Needs of Large Language Models. *arXiv preprint arXiv:2403.10081*, 2024.
30. Xinyu Wang, Yong Jiang, Nguyen Bach, Tao Wang, Zhongqiang Huang, Fei Huang, and Kewei Tu. RA-NER: Retrieval augmented NER for knowledge intensive named entity recognition. *Amazon Science Publications*, 2024.
31. Shengqiong Wu, Hao Fei, Leigang Qu, Wei Ji, and Tat-Seng Chua. Retrieval-augmented event argument extraction. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 2156–2168, 2023.
32. Kevin Wu, Eric Wu, and James Zou. How faithful are RAG models? Quantifying the tug-of-war between RAG and LLMs’ internal prior. *arXiv preprint arXiv:2404*, 2024.
33. Michihiro Yasunaga, Antoine Bosselut, Hongyu Ren, Xikun Zhang, Christopher D. Manning, Percy Liang, and Jure Leskovec. Deep bidirectional language-knowledge graph pretraining. In *Advances in Neural Information Processing Systems*, volume 35, pages 14951–14964, 2022.
34. Wenhao Yu, Dan Iter, Shuohang Wang, Yichong Xu, Mingxuan Ju, Soumya Sanyal, Chenguang Zhu, Michael Zeng, and Meng Jiang. Retrieval-augmented generation across heterogeneous knowledge. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 52–58, 2022.
35. In the real Dai, Chen Luo, Zhen Li, Xianfeng Tang, Hanqing Lu, Rahul Goutam, and Haiyang Zhang. RA-NER: Retrieval augmented NER for knowledge intensive named entity recognition. In *The Second Tiny Papers Track at ICLR 2024*, 2024.
36. Penghao Zhao, Hailin Zhang, Qinhan Yu, Zhengren Wang, Yunteng Geng, Fangcheng Fu, Ling Yang, Wentao Zhang, Jie Jiang, and Bin Cui. Retrieval-augmented generation for ai-generated content: A survey. *arXiv preprint arXiv:2402.19473*, 2024.

37. Bingfeng Zhou, Yangqiu Song, and Kam-Fai Wong. Event temporal relation extraction based on retrieval-augmented on llms. *arXiv preprint arXiv:2403.15273*, 2024.